Statistical Inference via Data Science: A ModernDive into R and the Tidyverse (Second Edition)

9 Hypothesis Testing

We have studied confidence intervals in Chapter 8. We now introduce hypothesis testing, another widely used method for statistical inference. A claim is made about a value or characteristic of the population and then a random sample is used to infer about the plausibility of this claim or hypothesis. For example, in Section 9.2, we use data collected from Spotify to investigate whether metal music is more popular than deep-house music.

Many of the relevant concepts, ideas, and we have already introduced many of the necessary concepts to understand hypothesis testing in Chapters 7 and 8. We can now expand further on these ideas and provide a general framework for understanding hypothesis tests. By understanding this general framework, you will be able to adapt it to many different scenarios.

The same can be said for confidence intervals. There was one general framework that applies to confidence intervals, and the `infer` package was designed around this framework. While the specifics may change slightly for different types of confidence intervals, the general framework stays the same.

We believe that this approach is better for long-term learning than focusing on specific details for specific confidence intervals. We prefer this approach also for hypothesis tests as well, but we will tie the ideas into the traditional theory-based methods as well for completeness.

In Section 9.1 we review confidence intervals and introduce hypothesis tests for one-sample problems; in particular, for the mean μ. We use both theory-based and simulation-based approaches, and we provide some justification why we consider it a better idea to carefully unpack the simulation-based approach for hypothesis testing in the context of two-sample problems. We also show the direct link between confidence intervals and hypothesis tests. In Section 9.2 we introduce the activity that motivates the simulation-based approach for two-sample problems, data collected from Spotify to investigate whether metal music is more popular than deep-house music. In Sections 9.3, 9.4, and 9.5 we explain, conduct, and interpret hypothesis tests, respectively, using the simulation-based approach of permutation. We introduce a case study in Section 9.6, and in Section 9.7 we conclude with a discussion of the theory-based approach for two-sample problems and some additional remarks.

If you'd like more practice or you are curious to see how this framework applies to different scenarios, you can find fully-worked out examples for many common hypothesis tests and their corresponding confidence intervals in the Appendices online.We recommend that you carefully review these examples as they also cover how the general frameworks apply to traditional theory-based methods like the t-test and normal-theory confidence intervals. You will see there that these traditional methods are just approximations for the computer-based methods we have been focusing on. However, they also require conditions to be met for their results to be valid. Computer-based methods using randomization, simulation, and bootstrapping have much fewer restrictions. Furthermore, they help develop your computational thinking, which is one big reason they are emphasized throughout this book.

Needed packages

If needed, read Section 1.3 for information on how to install and load R packages.

```
library(tidyverse)
library(moderndive)
library(infer)
library(nycflights23)
library(ggplot2movies)
```

Recall that loading the `tidyverse` package loads many packages that we have encountered earlier. For details refer to Section 4.4. The packages `moderndive` and `infer` contain functions and data frames that will be used in this chapter.

9.1 Tying confidence intervals to hypothesis testing

In Chapter 8, we used a random sample to construct an interval estimate of the population mean. When using the theory-based approach, we relied on the Central Limit Theorem to form these intervals and when using the simulation-based approach we do it, for example, using the bootstrap percentile method. Hypothesis testing takes advantages of similar tools but the nature and the goal of the problem are different. Still, there is a direct link between confidence intervals and hypothesis testing.

In this section, we first describe the one-sample hypothesis test for the population mean. We then establish the connection between confidence intervals and hypothesis test. This connection is direct when using the theory-based approach but requires careful consideration when using the simulation-based approach.

We proceed by describing hypothesis testing in the case of two-sample problems.

9.1.1 The one-sample hypothesis test for the population mean

Let's continue working with the population mean, μ. In Chapter 8, we used a random sample to construct a 95% confidence interval for μ.

When performing hypothesis testing, we test a claim about μ by collecting a random sample and using it to determine if the sample obtained is consistent with the claim made. To illustrate this idea we return to the chocolate-covered almonds activity. Assume that the almonds' company has stated on its website that the average weight of a chocolate-covered almond is exactly 3.6 grams. We are not so sure about this claim and as researchers believe that it is different than 3.6 grams on average. To test these competing claims, we again use the random sample `almonds_sample_100` from the `moderndive` package, and the first 10 lines are shown:

```
almonds_sample_100
```

```
# A tibble: 100 × 2
      ID weight
   <int>  <dbl>
 1   166    4.2
 2  1215    4.2
 3  1899    3.9
 4  1912    3.8
 5  4637    3.3
 6   511    3.5
 7   127    4
 8  4419    3.5
 9  4729    4.2
10  2574    4.1
# ℹ 90 more rows
```

The goal of hypothesis testing is to answer the question: "Assuming that this claim is true, how likely is it to observe a sample as extreme as or more extreme than the one we have observed?" When the answer to the question is: "If the claim was true, it would be very unlikely to observe a sample such as the one we have obtained" we would conclude that the claim cannot be true and we would reject it. Otherwise, we would fail to reject the claim.

The claim is a statement called the **null hypothesis**, H_0. It is a statement about μ, and it is initially assumed to be true. A competing statement called the **alternative hypothesis**, H_A, is also a statement about μ and contains all the possible values not included under the null hypothesis. In the almonds' activity the hypotheses are:

$$\begin{aligned} H_0: &\quad \mu = 3.6 \\ H_A: &\quad \mu \neq 3.6 \end{aligned}$$

Evidence against the null may appear if the estimate of μ in the random sample collected, the sample mean, is much greater or much less than the value of μ under the null hypothesis.

How do we determine which claim should be the null hypothesis and which one the alternative hypothesis? Always remember that the null hypothesis has a privileged status since we assume it to be true until we find evidence against it. We only rule in favor of the alternative hypothesis if we find evidence in the data to reject the null hypothesis. In this context, a researcher who wants to show some results or conclusions for new findings, needs to *prove* that the null hypothesis is not true by finding evidence against it. We often say that the researcher bears the *burden of proof*.

The hypothesis shown above represents a *two-sided test* because evidence against the null hypothesis could come from either direction (greater or less). Sometimes it is convenient to work with a *left-sided test*. In our example, the claim under the null hypothesis becomes: "the average weight is *at least* 3.6 grams" and the researcher's goal is to find evidence against this claim in favor of the competing claim "the weight is *less* than 3.6 grams." The competing hypotheses can now be written as $H_0 : \mu \geq 3.6$ versus $H_A : \mu < 3.6$ or even as $H_0 : \mu = 3.6$ versus $H_A : \mu < 3.6$. Notice that we can drop the inequality part from the null hypothesis. We find this simplification convenient as we focus on the equal part of the null hypothesis only and becomes clearer that evidence against the null hypothesis may come only with values to the left of 3.6, hence a left-sided test.

Similarly, a *right-sided test* can be stated $H_0 : \mu = 3.6$ versus $H_A : \mu > 3.6$. Claims under the null hypothesis for this type of test could be stated as "the average weight is *at most* 3.6 grams" or even "the average weight is *less* than 3.6 grams." But observe that *less* does not contain the *equal* part, how can the null then be $H_0 : \mu = 3.6$? The reason is related to the method used more than the semantics of the statement. Let's break this down: If, under the null hypothesis, the average weight is less than 3.6 grams, we can only find evidence against the null if we find sample means that are much greater than 3.6 grams, hence the alternative hypothesis is $H_A : \mu > 3.6$. Now, to find the evidence we are looking for, the methods we use require a point of reference, "less than 3.6" is not a fixed number since 2 is less than 3.6 but so too is 3.59. On the other hand, if we can find evidence that the average is greater than 3.6, then it is also true that the average is greater than 3.5, 2, or any other number less than 3.6. Thus, as a convention, we include the equal sign **always** in the statement under the null hypothesis.

Let's return to the test. We work with the two-sided test in what follows, but we will comment about the changes needed in the process if we are instead working with the left- or right-sided alternatives.

The theory-based approach

We use the theory-based approach to illustrate how a hypothesis test is conducted. We first calculate the sample mean and sample standard deviation from this sample:

```
almonds_sample_100 |>
  summarize(sample_mean = mean(weight),
            sample_sd = sd(weight))
```

```
# A tibble: 1 × 2
  sample_mean sample_sd
        <dbl>     <dbl>
1       3.682  0.362199
```

We recall that due to the Central Limit Theorem described in Subsection 7.3, the sample mean weight of almonds, $\overline{X}$, is approximately normally distributed with expected value equal to μ and standard deviation equal to $\sigma/\sqrt{n}$. Since the population standard deviation is unknown, we use the sample standard deviation, s, to calculate the standard error $s/\sqrt{n}$. As presented in Subsection 8.1.4, the t-test statistic

$$t = \frac{\overline{x} - \mu}{s/\sqrt{n}}$$

follows a t distribution with $n-1$ degrees of freedom. Of course, we do not know what μ is. But since we assume that the null hypothesis is true, we can use this value to obtain the test statistic as shown in this code next. Table 9.1 presents these values.

```
almonds_sample_100 |>
  summarize(x_bar = mean(weight),
            s = sd(weight),
            n = length(weight),
            t = (x_bar - 3.6)/(s/sqrt(n)))
```

x_bar	s	n	t
3.68	0.362	100	2.26

TABLE 9.1: Sample mean, standard deviation, size, and the t-test statistic

The value of $t = 2.26$ is the sample mean standardized such that the claim $\mu = 3.6$ grams corresponds to the center of the t distribution ($t = 0$), and the sample mean observed, $\overline{x} = 0.36$, corresponds to the t test statistic (t = 2.26).

Assuming that the null hypothesis is true ($\mu = 3.6$ grams) how likely is it to observe a sample as extreme as or more extreme than `almonds_sample_100` ? Or correspondingly, how likely is it to observe a sample mean as extreme as or more extreme than $\overline{x} = 0.36$? Or even, how likely is it to observe a test statistic that is $t = 2.26$ units or more away from the center of the t distribution?